5 **(a)** A copper wire of diameter 1.8 mm has a current of 17 mA flowing through it.

(i) The density of copper is 8.9×10^{3} kg m^{-3} and its molar mass is 63.5 g mol^{-1}.

Assuming that copper has one conduction electron per atom, show that the number density of charge carriers in the copper wire is 8.4×10^{28} m^{-3}.

[2]

(ii) Hence determine the drift velocity *v* of the charge carriers.

v = ... m s^{-1} [2]

(b) One segment of the copper wire in **(a)** is found to be thinner compared to the rest of the wire.

State and explain how the drift velocity of charge carriers will change when they flow into the thinner segment.

....

....

.... [2]

6 **(a)** State *Faraday's law of electromagnetic induction*.